Lightweight Heat Stress Detection for Cow Health Monitoring

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Abstract— **Farmers in the cattle industry stand to gain a great deal of profit and peace-of-mind from the monitoring services offered by companies such as Zalliant. Using numerous standalone mathematical and statistical AWS lambda functions, Zalliant analyzes streaming rumen temperature data to provide farmers with the necessary insights to make informed decisions about cow treatment plans that best suit their health needs. With the current landscape surrounding artificial intelligence and machine learning, Zalliant is interested in exploring the profitability of integrating these methods in their workflow. This project explores various supervised and unsupervised machine learning algorithms, including Long-Short-Term-Memory (LSTM) and logistic regression, comparing their performance and efficiency to Zalliant's current methods and other statistical approaches. This study finds that the nature of the problem is best suited for statistical detection methods, and while machine learning algorithms can reproduce the same results, the computational cost is not worth the minimal performance gains. Alternative statistical methods such as rolling window and STL decomposition outlier detectors provide lightweight and compact solutions capable of reducing the necessary sampling rate and may be a promising method to continue exploring for operation cost reduction.**

I. INTRODUCTION

Accounting for 22% of the agricultural sector in the United States, the cattle industry produces upwards of $100 billion annually [1]. Compared to other forms of livestock such as pork and poultry, the lifecycle of a harvest cow lasts approximately eight to twelve years, representing a significant investment for the farmer and ample time during which health complications which jeopardize the cows' profitability can occur. Diseases such as mastitis, for example, are difficult to track and can cost farmers hundreds of dollars per affected cow [2]. Companies such as Zalliant have acknowledged this problem and invested in research and development in monitoring methods that provide farmers with realtime insights and health information for their animals, enabling them to make informed decisions about how to best accommodate the health needs of the livestock [3].

Zalliant's health monitoring services are enabled by a proprietary device called a bolus, a compact internet-of-things (IoT) device which is swallowed by the animal and lives within the rumen, or the primary stomach of the cow, for the duration of the device's or the cow's lifespan. The device captures streaming temperature data at 5-minute intervals and sends the data to a server in 30-minute packets. Using these data, Zalliant uses a suite of mathematical and statistical Amazon Web Service (AWS) lambda functions to compute a baseline temperature and detect any anomalies that may be indicative of potential health risks. Farmers can access these anomalies via alerts published on Zalliant's proprietary dashboard.

The methodology and validation behind the current AWS lambda function suite is based in years of experience and comparison to live video feeds of the animals. They have gained a reliable reputation among their clients using their current methods, but they are always looking to improve the effectiveness and efficiency of their models. They have previously explored the use of machine learning (ML) and artificial intelligence (AI) tools in their workflow, but ultimately moved past the idea due to high operating costs for minimal performance improvements. In an age where AI and ML are growing rapidly and becoming more accessible, the applicability of these methods for Zalliant's purposes seem increasingly promising.

This project aims to evaluate the economic feasibility of integrating ML-based methods with Zalliant's current workflow for monitoring and detecting anomalies in streaming cow rumen temperature data. The study explores a wide range of both supervised and unsupervised ML baseline generation and anomaly detection algorithms, comparing them to Zalliant's existing methods and other statistical algorithms in terms of performance and efficiency.

Our findings suggest that, contrary to expectations, more complex machine learning approaches do not necessarily yield superior performance in this setting. Our proposed approach utilizes a lightweight, context-aware statistical framework based on rolling medians and robust deviation thresholds that prove to be more consistent, interpretable, and resilient to data gaps such as sensor issues and biological variability. This method achieves comparable or improved detection performance relative to Zalliant's existing system while identifying potential heat stress events significantly earlier by up to $\tilde{9}$ hours. Earlier detection of these events can enable more timely intervention, which can reduce treatment costs, improve animal welfare, and mitigate producitivity losses.

II. DATA

A. Dataset Description

The data used was accessed from a website called CowMonitor.com, Zalliant's dashboard for accessing the sensor data. From this platform, individual cow temperature records were selected and combined to create datasets for analysis. The data consists of continuous rumen temperature readings collected from Zalliant's bolus sensor, recorded at 5-minute intervals from two different farms: Grace-Way Farms, a dairy

Generative AI was referenced in this report for ideation and syntax assistance.

Variable	Description	Data Type
Timestamp	UTC time when measurement was recorded	DateTime
Temperature	Cow's internal rumen temperature (°C)	Float
Animal ID	Unique identifier for each cow	String/Integer
Device ID	Unique identifier for each bolus device	String/Integer
Alert	Flag indicating temperature-based event detected by Zalliant's system	Boolean
Connectivity Flag	Flag indicating connectivity issues between sensor and network	Boolean

TABLE I

DATASET VARIABLES AND THEIR DESCRIPTIONS

cow farm, located in New York, USA and Voltaurin, a beef cow farm, in South France.

For exploratory data analysis (EDA), a subset of 10 cows was constructed, including 5 dairy cows and 5 beef cows over a 7-day period. This subset supported initial exploration and visualization of the data to better understand temperature patterns. For model development and evaluation, a larger dataset of 30 cows was used, including 20 from the dairy cow farm and 10 from the beef cow farm over a 30-day period. The dataset was balanced to include an equal number of cows with and without alerts (alerts are temperature-based events that may indicate potential health issues) from Zalliant's existing temperature event detection methods, ensuring a fair comparison when evaluating detection performance.

The dataset includes several key variables used throughout the analysis, described in Table I.

B. Data Cleaning and Preprocessing

Temperature sensor data were analyzed per animal. Raw timestamps were parsed in UTC and merged with external alert annotations. Duplicate timestamps were removed and temperature streams were sorted chronologically.

Timestamp intervals were confirmed at 5-minute intervals with minimal transmission gaps.

1) Blue Line Replication — Baseline Temperature Reconstruction: The blue line replication method reconstructs each animal's baseline temperature profile by systematically identifying and removing drinking-induced artifacts from the raw sensor data.

For each animal, temperature values were sorted by timestamp and indexed for rolling computation. A 1-day rolling mode served as the local baseline temperature, representing the most frequent temperature value within each rolling window. Deviation from baseline was computed as:

$$\text{difference} = \text{temperature} - \text{rolling mode} \tag{1}$$

Only negative deviations (values below baseline) were considered for bout detection. Within each rolling window, the first quartile (Q_1), third quartile (Q_3), and interquartile range ($\text{IQR} = Q_3 - Q_1$) of negative differences were computed. A dynamic lower threshold was defined as:

$$\text{lower bound} = Q_1 - \text{IQR\_MULT} \times \text{IQR} \tag{2}$$

A bout was flagged when:

$$\text{difference} < \text{lower bound} \tag{3}$$

This IQR-based approach adaptively identifies sharp, short-duration temperature drops characteristic of drinking behavior. Identified bout periods were excluded from subsequent analysis. The cleaned temperature series was resampled to a fixed 15-minute grid, with bin values computed as the mean within each interval. Missing bins were filled by linear interpolation, producing a uniform time series suitable for rolling-window statistical operations.

To limit artifacts from sparse sampling, a suppression mask was applied: gaps longer than 3 hours were masked, and the first 7 days of each animal's record were suppressed to ensure sufficient historical data for baseline estimation.

The output—the blue line—represents the reconstructed baseline temperature trajectory free from drinking-related distortions, serving as input for event detection.

C. Exploratory Data Analysis

Exploratory analysis of the temperature data reveals several consistent patterns across cows. The overall temperature distribution is centered around approximately 38.5°C and exhibits a slight left skew, driven by frequent sudden drops observed in the time series.

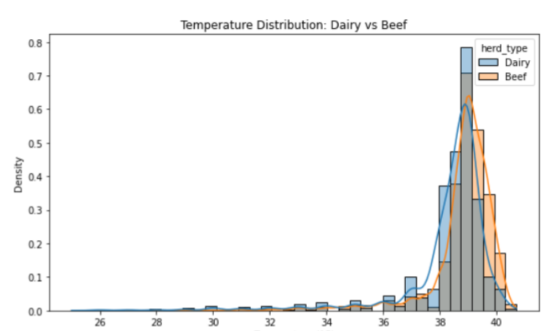

Fig. 1. Distribution of rumen temperature across dairy and beef cows.

These drops occur regularly and introduce substantial variability in the data. Further investigation suggests that many of these short-term temperature decreases are behavior-driven rather than indicative of true health-related events. In particular, drinking activity can cause temporary reductions in rumen temperature, leading to sharp but transient drops.

As a result, not all observed decreases in temperature should be treated as abnormal events.

Comparison across herd types shows that dairy and beef cows share similar overall temperature distributions, though differences in variability are observed. Dairy cows tend to exhibit greater daily temperature fluctuations, while beef cows display slightly more pronounced left-skewed distributions, suggesting a higher frequency of transient drops.

Analysis of the alert variable further highlights the challenges in event detection. Alerts represent only a small fraction of the dataset and are associated with significantly lower temperature values (Fig 2) compared to normal readings. This indicates that the existing detection system primarily captures more extreme cases, while more subtle deviations remain difficult to identify.

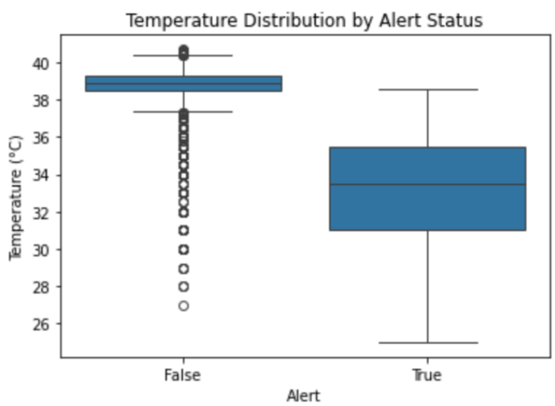

Fig. 2. Temperature distribution by alert status.

Overall, these observations highlight the challenge of distinguishing true abnormal temperature events from normal behavioral fluctuations, motivating the need for approaches that incorporate both temporal patterns and baseline-relative deviations.

III. Model Development

A. Context-Aware Event Detection

With alerts representing only a small fraction of the dataset, we treated event detection as a time-indexed univariate unsupervised problem where temperature was the sole available feature. Operational constraints—deployment on a single GPU with approximately 15 seconds inference time per cow—necessitated lightweight methods rather than complex deep learning models.

Our approach employs a robust statistical detector designed to identify sustained temperature rises using dual rolling windows for baseline estimation and adaptive scaling.

1) Long-term baseline and deviation score: Our goal is to identify *unusual sustained increases* in temperature for each animal, not simply high temperatures in absolute terms. Different animals can have different typical temperature levels, and even the same animal can vary over time. Therefore, instead of modeling raw temperature directly, we model *deviation from that animal's recent normal pattern*.

We first estimate each animal's long-term normal temperature (baseline) using a 7-day rolling median:

$$b_t = \mathrm{median}(T_{t-7d:t}) \tag{4}$$

where T_t is the resampled temperature and b_t is the baseline at time t.

We then compute residuals (deviations from normal):

$$r_t = T_t - b_t \tag{5}$$

These residuals form a residual distribution over time. Most residuals represent normal fluctuation around baseline, while only the tail values represent unusual behavior. Our detector is designed to identify these extreme positive residuals (unusually high deviations).

a) Why a 7-day baseline window?: A 7-day window captures slow physiological drift and day-to-day variation, while smoothing short transient spikes. We use the median (rather than mean) because it is less sensitive to outliers and occasional noisy measurements.

To compare residual magnitude fairly across animals and time periods, we robustly standardize residuals with a MAD-based z-score:

$$z_t = \frac{r_t - \tilde{r}_t}{1.4826 \cdot \max(\mathrm{MAD}_t, 0.1) + \epsilon} \tag{6}$$

where $\tilde{r}_t$ is a rolling median of residuals and MAD_t is the rolling median absolute deviation:

$$\mathrm{MAD}_t = \mathrm{median}\left(|r - \tilde{r}_t|\right) \tag{7}$$

Here, MAD (median absolute deviation) is a robust measure of spread: it quantifies typical variability while being less affected by extreme values than variance or standard deviation. The factor 1.4826 scales MAD to be comparable to standard-deviation-based z-scores under Gaussian assumptions. The floor term $\max(\mathrm{MAD}_t, 0.1)$ prevents unstable score inflation when local variability is extremely small.

b) Why a separate 14-day scale window?: We use a second, longer window (14 days) for scale estimation ($\tilde{r}_t$, MAD_t) because the detector is built on residuals, and residuals computed within only the 7-day baseline horizon can still contain local tail values that appear anomalous in that short context. To avoid treating these short-horizon extremes as the reference for "normal" spread, we estimate scale from a broader history. In effect, the current residual is evaluated against a larger distribution of past residual behavior, which yields a more stable denominator and reduces volatility in the anomaly score.

2) Adaptive thresholding: After converting residuals into robust z-scores, we still need to decide what is "unusually high" for each animal. A single global threshold is often unreliable because animals differ in baseline temperature patterns and natural variability. A score that is extreme for one animal may be typical for another.

To address this, we estimate an animal-specific threshold from that animal's own score distribution. In practice, we

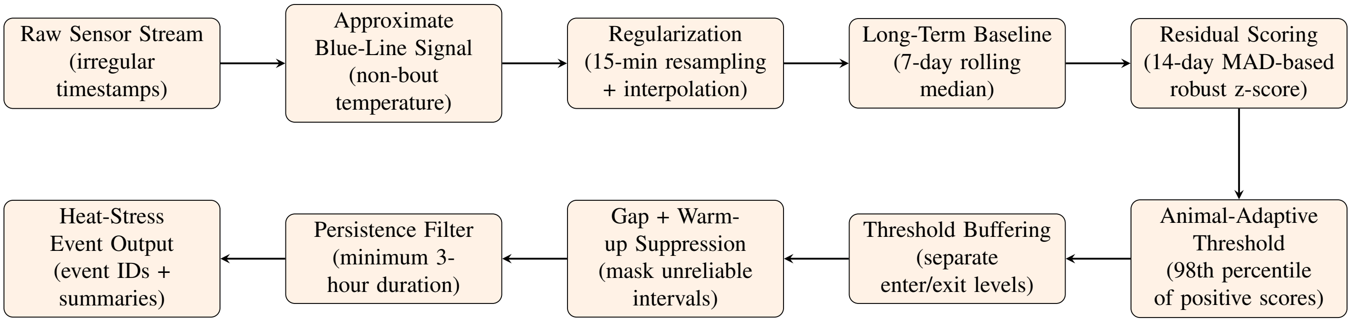

Fig. 3. Context-aware statistical pipeline used for long-term heat-stress event detection.

focus only on positive scores (upward deviations) and take the 98th percentile:

$$\theta_{\text{animal}} = P_{98}\big(\max(z_t, 0)\big) \tag{8}$$

Intuitively, this means that only the top tail of each animal's positive deviation distribution is considered candidate abnormal behavior. This keeps sensitivity personalized while avoiding manual per-animal tuning.

3) Threshold buffering and event direction: Even with a good threshold, scores near the boundary can fluctuate above and below the cutoff from one time step to the next. If a single threshold is used for both start and stop decisions, this can cause unstable alert toggling.

We therefore use *threshold buffering*: one threshold to enter an event and a lower threshold to exit it:

$$\theta_{\text{enter}} = 1.00 \cdot \theta_{\text{animal}}, \qquad \theta_{\text{exit}} = 0.75 \cdot \theta_{\text{animal}} \tag{9}$$

The event state is updated as:

$$\text{enter event if } z_t \geq \theta_{\text{enter}}, \qquad \text{exit event if } z_t < \theta_{\text{exit}} \tag{10}$$

This buffered rule stabilizes event boundaries by requiring a clearer drop before ending an event.

Our detection target is high-temperature risk; therefore, only positive deviations are considered for triggering. Negative deviations are clipped and do not initiate events.

4) Gap suppression: Our detector assumes that the temperature timeline is sufficiently continuous. When there are long data gaps, interpolated values in those regions are less trustworthy, and alerts triggered there may be misleading. To prevent this, we apply *gap suppression*: if consecutive cleaned readings are separated by more than 3 hours, the corresponding gap interval is masked and no alerts are emitted in that period.

The same masking is also used when estimating thresholds, so uncertain intervals do not influence what is considered "normal" or "extreme." We additionally allow an optional initial warm-up period at the start of each series, during which alerts are temporarily suppressed so rolling baseline and variability estimates can stabilize.

5) Persistence filtering and event construction: A single high score at one time point is often not clinically meaningful; it may reflect transient behavior or measurement noise. We therefore require *persistence*: a candidate anomaly is retained only if it remains active for at least 3 hours (about 12 consecutive bins on a 15-minute grid).

This step shifts the detector from point-wise spikes to sustained episodes, which better matches the intended use case of prolonged temperature elevation. After persistence filtering, adjacent positive bins are merged into one continuous event and assigned a unique event ID.

For each event, we report interpretable summaries, including start and end time, duration, temperature statistics (mean/max/min), and score statistics (e.g., mean and peak event score).

Figure 4 illustrates the final event-detection output after threshold buffering, suppression, and persistence filtering.

6) Key Enhancements and Design Rationale: The detection pipeline incorporates several key enhancements that improve upon simpler rolling window approaches. Consecutive anomalous bins are grouped into discrete events rather than treating each time point independently, reducing fragmentation and providing clinically meaningful event boundaries. Gap suppression removes false signals arising from missing data or sensor connectivity issues, preventing artifacts from triggering spurious alerts.

Compared to simple quarterly rolling baselines, the dual-window approach—combining a 7-day rolling median for baseline estimation with a 14-day MAD for robust scaling—enables earlier detection of heat stress events by adapting more quickly to local temperature trends while maintaining robustness to short-term fluctuations.

A critical design decision was to focus exclusively on detecting temperature rises rather than drops. Since temperature decreases are highly correlated with drinking behavior, and our bout detection method is an approximation that may not capture all drinking artifacts, detecting drops would introduce systematic bias. By clipping negative deviations to zero and applying positive-only hysteresis, the method avoids false positives from residual drinking-related noise while maintaining high sensitivity to sustained elevations indicative of health events.

7) External Alert Comparison: For each animal, we compared the earliest model-detected event start time with the earliest external alert timestamp. External alerts falling within model-suppressed intervals were excluded to avoid penalizing intentionally masked regions.

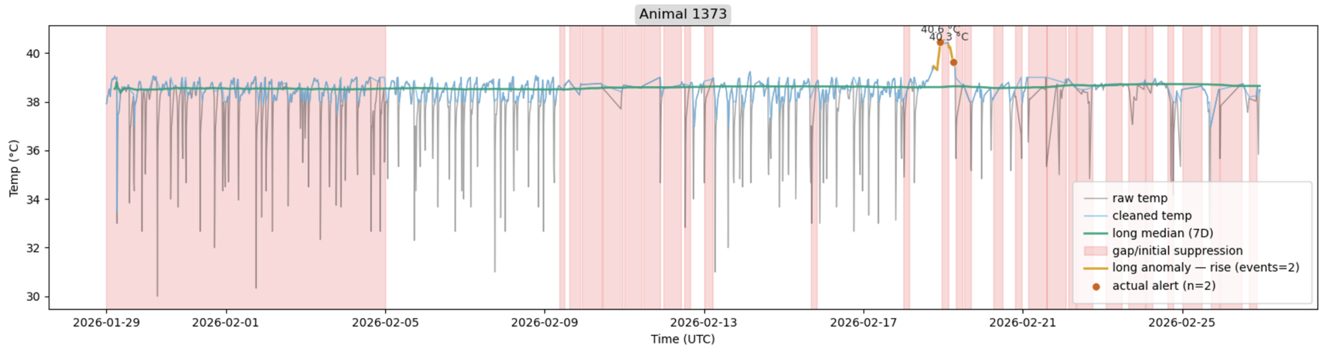

Fig. 4. Example model output on cleaned and resampled temperature data, showing detected heat-stress events.

B. Alternative Approaches

Several supervised and deep learning approaches were explored before adopting the statistical method.

1) Logistic Regression: Logistic regression was applied with engineered temporal features including lags, rolling statistics, and temperature changes. However, this approach required labeled ground truth data, which was limited in the dataset. Performance varied significantly across individual cows, and the model was sensitive to class imbalance, making it unreliable for consistent detection.

2) XGBoost: A gradient boosting classifier was trained to predict short-term alert signals, achieving ROC AUC = 0.70, precision = 0.52, and recall = 0.53. Despite moderate performance on aggregate metrics, predictions were noisy and unstable, rendering the model unsuitable for real-time event triggering. Like logistic regression, XGBoost's reliance on limited labeled data constrained its effectiveness.

3) LSTM Networks: Long Short-Term Memory (LSTM) networks were implemented to model temperature as a sequential time series, learning temporal patterns from past readings to predict future alert events. The approach yielded moderate and inconsistent performance (F1 = 0.53) across animals. Additionally, LSTM models were computationally expensive, with inference times exceeding operational constraints for deployment on a single GPU.

IV. Results and Discussion

A. Model Performance

The model was evaluated on a test set of 30 cows: 20 from dairy herds and 10 from beef herds, with a balanced 50:50 split between cows with existing alerts and those without. Performance was assessed by comparing model-detected events against Zalliant's current alert system.

Fig. 5 presents the confusion matrix comparing model predictions with actual alerts. The model achieved strong performance across standard classification metrics: Accuracy = 0.80, Precision = 0.747, Recall = 0.867, and F1 Score = 0.813.

The high recall (0.867) indicates the model successfully identified 86.7% of events flagged by the current system, while precision (0.747) suggests that 74.7% of model-detected events corresponded to actual alerts. The model

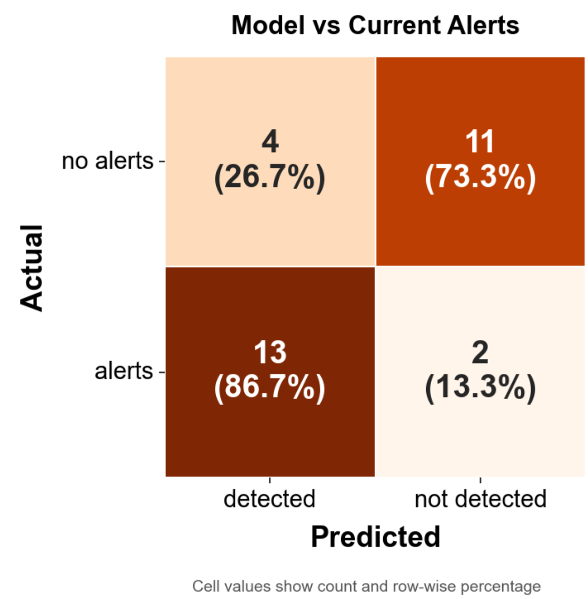

Fig. 5. Confusion matrix comparing model-detected events with current alert system.

detected 13 out of 15 true alert cases, with only 2 false negatives, while generating 11 new detections not present in the current system.

1) Early Detection Performance: A critical advantage of the proposed method is its ability to detect temperature events earlier than the current system. Model alerts were matched with Zalliant's alerts by filtering for high-confidence events and linking only temporally proximate alerts as corresponding to the same physiological event. The model achieved a median early detection time of 9 hours ahead of current alerts, providing farmers with earlier intervention opportunities.

2) Robustness and Adaptability: The model demonstrates robustness to noise through multi-stage filtering: bout removal, rolling baseline estimation, robust scaling via MAD, and persistence filtering. By computing animal-specific empirical thresholds rather than applying global cutoffs, the method adapts to individual cow variation and seasonal temperature patterns, enabling detection of more physiological signals while maintaining low false positive rates. "'

B. Economic Impact

Overall, our model offers significant potential for cost savings, both to Zalliant and to the client farmers. For Zalliant, the model retains solid performance at reduced sampling rates, increasing the battery life of the small bolus and reducing hardware costs related to bolus replacement by up to 50%. Current methods operate using streaming data recorded at a 5 minute sampling interval, but our event detection model can double, or potentially triple this rate while maintaining relatively similar, if not improved performance.

TABLE II
MODEL PERFORMANCE METRICS BY SAMPLING RATE

Sampling Rate (min)	Precision	Recall	Specificity	Accuracy
5 (baseline)	0.86	0.75	0.92	0.85
10	0.88	0.88	0.92	0.90
15	0.70	0.88	0.75	0.80

The heat stress detection models also outperform current methods by detecting high temperature events up to 9 hours earlier. The key benefit of products like Zalliant's is the advantage of detecting health anomalies earlier than visual inspection. Thus, it follows that any method that enables earlier detection is an improvement. Just hours of advance detection can save farmers in health expenses, up to an estimated $10K per 100 cows annually for detection 9 hours earlier.

C. Conclusion

The objective of this project was to identify an operationally optimal temperature-event detector: one that performs at least as well as the current approach while reducing computational cost.

Under this objective, supervised ML methods were limited by sparse labels and class imbalance, which reduced reliability across animals. Deep sequence models (e.g., LSTM) showed anomaly-detection potential but introduced higher inference cost and inconsistent event-level behavior for the target deployment setting.

The context-aware event detection model provided the best practical trade-off for current needs: comparable (and in some settings improved) detection behavior relative to existing alerts, stronger operational stability, and substantially lower computational burden.

D. Suggestions for Future Work

Distortions in the baseline temperature caused by drinking water bouts made low temperature events extremely difficult to predict, resulting in the decision to focus entirely on high temperature events instead. Although the blue line replication was meant to reduce the effects of the temperature drops, the methods were not enough to enable reliable low temperature event detection. In the future, additional methods can be explored to apply similar lightweight methods to low temperature event detection.

Throughout the course of model development, the most significant obstacle was the issue of limited labeled data. On average, each cow experienced less than one labeled alert per month, making supervised learning methods almost entirely infeasible. Additionally, the labeled datapoints were initially defined by Zalliant's current methods, disconnecting model predictions from ground truth by another degree of separation. In the future, it may be worth looking into establishing more frequent, tangible events or target labels to predict using lightweight supervised learning methods.

The input variable stream being limited to only temperature was also a concern throughout the project. As a time series, there was minimal noticeable variation from routine behavior, making event detection difficult. Predictive capabilities can be enhanced in the future by expanding the input stream to additional variables. Data such as drinking water temperature, or other biometrics such as heart rate should be explored to improve predictive capability across all models.

V. ACKNOWLEDGMENTS

We would like to thank the team at Zalliant: John Balbian, Utsav Uts, and Emmanual Mounier for their patience and support. We would also like to thank our course instructors, Professor Anand and Professor Caliskan, for their help and guidance throughout the semester.

REFERENCES

[1] R. Knight, "Cattle & beef – sector at a glance," https://www.ers.usda.gov/topics/animal-products/cattle-beef/sector-at-a-glance, 2026, accessed: 2026-03-17.

[2] E. Rollin, K. C. Dhuyvetter, and M. W. Overton, "The cost of clinical mastitis in the first 30 days of lactation: An economic modeling tool," *Preventive Veterinary Medicine*, vol. 122, no. 3, pp. 257–264, 2015, pMID: 26596651.

[3] J. L. Kleen and R. Guatteo, "Precision livestock farming: What does it contain and what are the perspectives?" *Animals*, vol. 13, no. 5, p. 779, 2023.